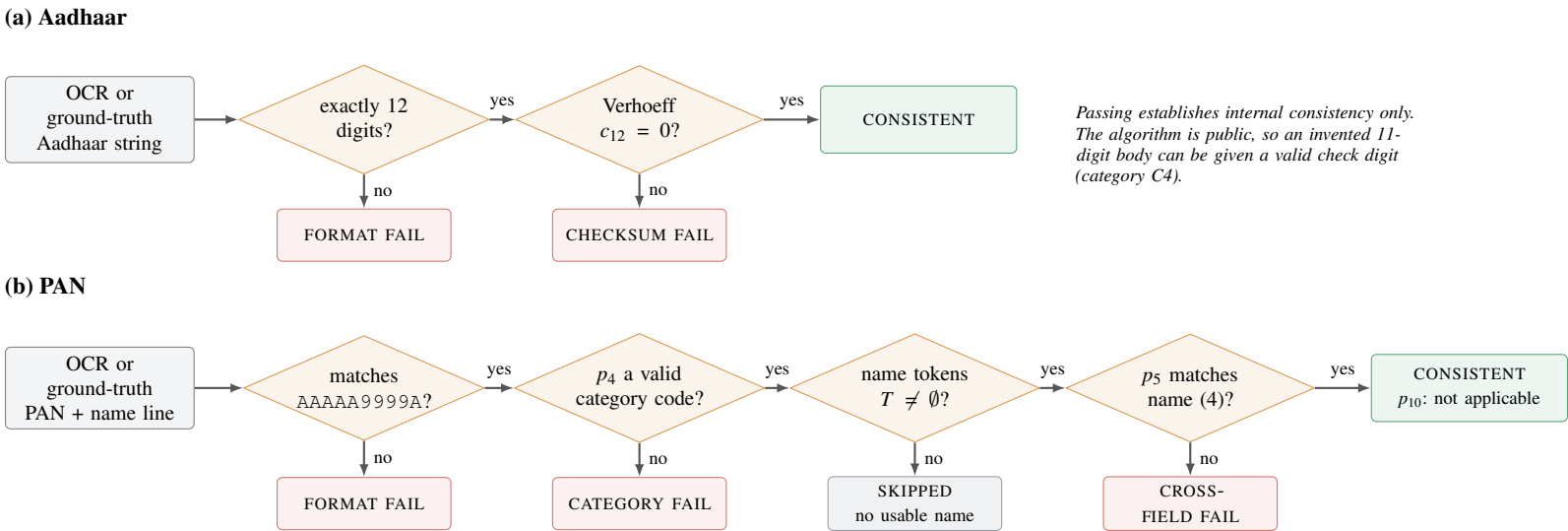
(a) Aadhaar
OCR or ground-truth Aadhaar string
exactly 12 digits?
yes
Verhoeff $c_{12} = 0$?
yes
CONSISTENT
no
FORMAT FAIL
no
CHECKSUM FAIL
Passing establishes internal consistency only. The algorithm is public, so an invented 11-digit body can be given a valid check digit (category C4).
(b) PAN
OCR or ground-truth PAN + name line
matches AAAAA9999A?
yes
p_4 a valid category code?
yes
name tokens $T \neq \emptyset$?
yes
p_5 matches name (4)?
yes
CONSISTENT
p_{10}: not applicable
no
FORMAT FAIL
no
CATEGORY FAIL
no
SKIPPED no usable name
no
CROSS-FIELD FAIL